38.

- An estuary is a place where a river or a stream opens into the sea (mouth of the river).
- It is a partially enclosed coastal area of brackish water (salinity varies between 0-35 ppt) with one or more rivers or streams flowing into it, and with a free connection to the open sea.
- At the estuaries, freshwater carrying fertile silt and runoff from the land mixes with the salty sea water.
- Estuaries form a transition zone (ecotone) between river environments and maritime environments.
- Examples of estuaries are river mouths, coastal bays, tidal marshes, lagoons and deltas.
- Estuaries are formed due to rise in sea level, movement of sand and sandbars, glacial processes and tectonic processes.
- All the plants and animals in the estuaries are subjected to variations in salinity to which they are adapted (osmoregulation).
- Estuaries are greatly influenced by tidal action. They are periodically washed by sea water once or twice a day based on the number of tides.
- In some narrow estuaries, tidal bores are significant. Tidal bores cause great damage to the estuarine ecology.

39. According to their existence in nature:

- Quantitative Pollutants: These occur in nature and become pollutant when their concentration reaches beyond a threshold level. For example, carbon dioxide, nitrogen oxide.
- Qualitative Pollutants: These do not occur in nature and are human-made. For example, fungicides, herbicides, DDT, etc.

Composition

- Fly ash particles are oxide rich and consist of silica, alumina, oxides of iron, calcium, and magnesium and toxic heavy metals like lead, arsenic, cobalt, and copper.
- Major oxides are present are aluminium silicate (in large amounts), silicon dioxide (SiO_2) and calcium oxide (CaO).

40.

- Up to about 70% of light absorption is by black carbon. When Brown carbon is present independently it has nearly 15% potential to warm the atmosphere by absorbing light.